Australia Awards

REAP Form
(for Applicants of 2026 Intake)

The Re-entry Action Plan (REAP) is a formal document that represents a scholar's commitment to honor the conditions of the Australia Awards Scholarships (see Scholarship Handbook section 4.2). It is a mechanism that allows scholars to apply what they have learned in Australia to contribute to Philippine development through a meaningful project in an organisation or community.

I. RELEVANCE OF THE REAP. The REAP should clearly and succinctly explain how the delivery of its proposed output or outputs will improve the services or performance of the Host Organisation and ultimately contribute to the achievement of the goals set in the Philippine Government's Ambisyon Natin 2040, and the goals of the Australian Government in the Philippines. Fill out the following sections after reading these required documents.	
Name of Proponent	Rhodolph C. Nullar
REAP Title	**PRIgpt: RAILytics** – A Localized AI Tool for Automating Competency Gap Analysis for the Metro Rail Transit Line 3 (MRT-3) Human Resources Using Large Language Model (LLM)-Powered Data Analysis
REAP Objective	The objectives of the REAP are divided into three sections, namely: 1. Develop a comprehensive competency database of RO personnel by gathering and organizing data on their work experience, educational background, trainings and seminars attended, professional certifications, and other relevant details needed for the competency gap analysis, to be completed within 10 months. 2. Design the PRIgpt: RAILytics (Rail Analytics) tool by analyzing the large datasets from the competency database and teaching the LLM how to examine and study available data based on the Philippine Skills Framework for Railway Operations and Maintenance (O&M) personnel, to be completed within 14 months. 3. Develop the PRIgpt: RAILytics Guideline which outlines how the automation process was accomplished, from data collection to competency gap analysis, to be completed within 24 months. This guideline may also serve as a reference for the

	recommendation of future PRI training courses for the development of human resources in the railway sector.
Name of Host Organisation	Department of Transportation – Philippine Railways Institute
Description of the Host Organisation and relation to REAP	As an instructor in PRI's Training Division (TD), I have been actively involved in developing and delivering training programs for railway personnel, particularly in Refresher Training (RT), Fundamental Training (FT), Capacity development training (CDT) courses. My work has provided firsthand insights into the challenges of competency management and workforce development, which led me to choose PRI as my host organization for implementing my Re-Entry Action Plan (REAP). PRI serves as the primary training and research arm for railway personnel in the Philippines, making it the ideal platform to develop PRIgpt: RAILytics—an AI-driven competency gap analysis and training recommendation tool. Working at DOTr-PRI will be instrumental in the successful implementation of my REAP. Being part with the Training Division, I have direct access to training data, key stakeholders, and decision-makers, which will facilitate data gathering, system integration, and policy adoption. This strong connection ensures that PRIgpt: RAILytics will be aligned with PRI's long-term training goals, ensuring its sustainability and scalability.
BACKGROUND INFORMATION	
Problem / Opportunity	One of the critical yet overlooked challenge in the railway sector is the lack of a structured competency database and an automated competency gap analysis for railway personnel. Currently, PRI conducts various training programs but there is no centralized, structured repository of railway personnel competencies, work credentials, and training records. This comprehensive database will be crucial as the information will be used as a foundational reference to gain an overview of the competency profile of human resources in the railway sector. In contrast, other organizations, such as Singapore's Land Transport Authority (LTA), have implemented structured skills databases that ensure targeted training interventions for railway professionals. Another key issue is that competency gap analysis at PRI remains largely manual and subjective. Without a structured competency database, identifying training needs is based on managerial perception rather than data-driven insights. This reactive approach results in personnel receiving training that may not fully address their competency deficiencies. In comparison, the European Railway Agency (ERA) uses AI-powered workforce planning tools to map competency gaps and optimize training programs, efficiency, and workforce retention. Additionally, PRI lacks automation in its training recommendation process. The institute currently relies on manual assessments and traditional HR practices, which are time-consuming and inconsistent. This leads to delays in identifying critical skill development programs, which can negatively impact railway operations. Moreover, resources may be used inefficiently, as

	some RO personnel may receive training they are already good at, while others with critical skill gaps remain untrained. In contrast, the UK's National Skills Academy for Rail (NSAR) integrates AI-based analytics to provide personalized training pathways, ensuring each railway worker receives the most relevant training based on real-time competency assessments. By implementing PRIgpt: RAILytics, PRI can transition from a manual and subjective training needs assessment to an automated, AI-driven system that ensures data-driven competency gap analysis, improving the accuracy of training recommendations. This system will also optimize resource allocation by reducing unnecessary or misaligned training expenditures and improve workforce competency tracking. Ultimately, this REAP presents an opportunity to modernize PRI's training framework while laying the foundation for nationwide adoption once a Philippine railway skills framework is established.
BENEFICIARIES	
Direct Beneficiary	The immediate beneficiaries of the REAP will be the whole PRI organization, particularly those involved in workforce development. By implementing a comprehensive competency database and the PRIgpt: RAILytics tool, trainers, curriculum developers, and policymakers will gain data-driven insights into competency gaps of RO personnel, enabling us to design more targeted and effective training programs. Trainers and curriculum developers will gain real-time insights into training needs, allowing them to design more relevant and effective courses. Furthermore, PRI top management and policymakers will be able to improve resource allocation and policy decisions to support long-term railway sector development. RO personnel will also benefit from a structured competency database that provides a clear assessment of their skills, training history, and certification requirements. Through PRIgpt: RAILytics, they will receive targeted training recommendations based on identified competency gaps, ensuring continuous professional development and career growth. This data-driven approach will help ROs optimize workforce planning, enhance operational efficiency, and align personnel competencies with industry standards.
Gender Equality, Disability, and Social Inclusion (GEDSI)	The implementation of the proposed REAP will promote inclusive workforce development within the Philippine railway sector by ensuring that women and marginalized groups, including Indigenous Peoples (IP), Persons with Disabilities (PWDs), and LGBTQ++ individuals, have equal access to training opportunities and career advancement. By leveraging data-driven insights from PRIgpt: RAILytics, the system can help identify disparities in training participation, career progression, and competency development among these groups, enabling targeted interventions to promote inclusivity. To achieve this, the competency database will track the participation of women and marginalized groups in training programs, ensuring that PRI training modules incorporate gender-sensitive and diversity-inclusive curricula that foster fair opportunities in both technical and leadership roles. IP and

	LGBTQ++ individuals will benefit from career development programs tailored to their unique needs, leveraging PRIgpt: RAILytics to identify competency gaps and offer targeted training recommendations. By implementing these strategies, the REAP will help reduce barriers to career growth and ensure that all railway personnel, regardless of gender, ability, or background, have equal opportunities for training, upskilling, and professional advancement in the railway industry.
Climate Change Adaptation, Disaster Risk Reduction and Environmental Protection	Through PRIgpt: RAILytics, the project will help optimize competency-based training for railway personnel, ensuring they are well-equipped to operate and maintain an efficient, well-managed railway system. This supports the shift toward sustainable public transportation, reducing dependence on carbon-intensive modes of travel and lowering greenhouse gas emissions. In terms of implementation, the digitalization of competency management will significantly reduce the need for paper-based records and manual documentation, minimizing waste and promoting eco-friendly practices within DOTr-PRI. Additionally, by leveraging AI and automation, the initiative reduces redundant training sessions and optimizes resource allocation, lowering energy consumption and operational inefficiencies. By integrating competency-driven insights into railway training, the project strengthens transport resilience, helping the railway sector adapt to climate risks while promoting sustainable and environmentally responsible railway operations.

II.	**IMPACT AND SUSTAINABILITY OF THE ORGANISATIONAL OUTCOME OF THE REAP.** The REAP should produce an output/s that will help the Host Organisation improve their services or performance. These improvements are the outcome/s of your REAP. You should therefore clearly articulate your strategies for the Host Organisation will sustain the gains of the REAP after its completion. (TIP: Based on our survey, the following factors ensure the effective implementation of REAP: the goals of the REAP should align with the goals of the Host Organisation and the Host Organisation is committed to the completion and full implementation of your REAP.)
Development impact and Sustainability	The proposed REAP aligns with the objectives outlined in the Sustainable Development Goals (SDGs), Philippine Development Plan (PDP) 2023-2028, AmBisyon Natin 2040, and Pagtanaw 2050, particularly in enhancing transportation infrastructure and human capital development. This project also contributes significantly to the priority focus areas of the DOTr *which include digital transformation initiatives and* human resource management and development. This REAP contributes to Sustainable Development Goal (SDG) 4 (Quality Education), specifically on Target 4.4, by enhancing competency-based training for railway personnel, ensuring continuous skills development and alignment with industry standards. Additionally, it supports SDG 9 (Industry, Innovation, and Infrastructure) by integrating AI-driven workforce analytics

	into railway operations, promoting digital transformation and sustainable infrastructure development in the transportation sector. The PDP 2023-2028 emphasizes the need to expand and upgrade infrastructure, focusing on delivering sustainable, resilient, integrated, and modern systems. This includes the development of mass transportation systems like railways to accommodate increasing demand and improve accessibility and safety for all users. The REAP's initiative to develop a comprehensive competency database and implement AI-driven analytics for railway personnel directly supports these goals by ensuring that the workforce is adequately trained to operate and maintain enhanced railway systems effectively. The REAP complements these strategic objectives by promoting systematic workforce development in the railway sector, which is essential for the successful implementation of the country's transportation infrastructure initiatives. The REAP's outputs will contribute to the achievement of the AmBisyon Natin 2040 vision, particularly in the "Maginhawa" vision element, which ensures the comfortability and security of Filipino's lives through good transport facilities. The integration of AI-powered workforce planning in railway training contributes to the modernization and digital transformation of public transport, aligning with the long-term goal of providing a more accessible, convenient, and sustainable transportation system for all Filipinos. The REAP aligns with Pagtanaw 2050 by leveraging AI-driven competency management to enhance the digital transformation and sustainability of the Philippine railway sector. By optimizing workforce skills development through data analytics and automation, the project supports science, technology, and innovation (STI)-driven governance, which is a key pillar of Pagtanaw 2050's vision for a future-ready Philippines. To ensure the institutionalization and sustainability of the proposed REAP after its implementation, I will establish a multi-faceted strategy focusing on policy integration, capacity-building, stakeholder engagement, and long-term system scalability within the PRI organization. First, I will work towards policy integration by advocating for the official adoption of the competency database and PRIgpt: RAILytics into PRI's training and training development framework. This includes collaborating with PRI top management and HR teams to institutionalize data-driven decision-making for competency assessments, and training needs analysis. The PRIgpt: RAILytics Guideline will serve as a reference document to formalize the automation process within PRI's policies and standard operating procedures. Second, capacity-building will be a critical component to ensure the sustainability of PRIgpt: RAILytics. I will facilitate training sessions and knowledge transfer programs for PRI personnel, particularly HR and training staff, equipping them with the necessary skills to maintain and improve the system over time. Lastly, scalability and continuous improvement will be embedded into the REAP's implementation. While the initial scope focuses on the human resources of one RO, the long-term plan includes

	expanding PRIgpt: RAILytics to other ROs. The system will be designed in a modular and adaptable manner to integrate new data sources and accommodate evolving workforce needs. Additionally, periodic performance evaluations and impact assessments will be conducted to refine and optimize the tool's effectiveness.
Organisational Outcome	1. Enhancement of Service Delivery The development of PRIgpt: RAILytics will improve how PRI identifies training needs, ensuring that RO personnel receive data-driven, targeted, and timely training interventions. This will enhance efficiency in workforce development and ultimately contribute to better-trained railway personnel, leading to improved railway operations and safety standards. 2. System Development This proposed REAP introduces a comprehensive competency database and an AI-powered competency gap analysis system. These innovations will establish a structured and automated system for PRI to manage RO personnel competencies, replacing traditional and obsolete manual methods. 3. Policy Evaluation and Formulation By analyzing competency data, PRIgpt: RAILytics can help assess the effectiveness of existing training policies and provide evidence-based recommendations for policy updates related to railway training programs. This can guide future policy formulation within the Department of Transportation and the broader railway sector. 4. Building Competencies By systematically identifying skill gaps and recommending relevant training programs, this proposed REAP will help PRI enhance the competencies of railway personnel. This ensures that PRI's workforce remains aligned with the competency-based framework, improving the overall skill level and career progression opportunities for railway employees.
Output	The proposed REAP will generate three key outputs: 1. A Comprehensive Competency Database of PRI Personnel This database will serve as the foundation for workforce planning by consolidating personnel data on professional credentials, education, training, and work experience. This will allow PRI to identify competency gaps systematically and develop more targeted training programs. 2. The PRIgpt: Railytics Tool for Competency Data Analysis The PRIgpt: RAILytics tool will leverage artificial intelligence to analyze these datasets and provide data-driven insights for decision-making, ensuring that training programs align with Philippine skills framework for railway O&M personnel. 3. The PRIgpt: RAILytics Guideline This guideline will document the entire automation process and training recommendations, serving as a reference for scaling the system and informing future policy and training strategies for all railway personnel. These outputs will help PRI enhance training effectiveness, workforce competency, and strategic planning. By implementing data-driven workforce development, the REAP will contribute to the modernization and efficiency of railway training, supporting

	PRI's long-term goal of strengthening human capital in the railway sector.
Competencies	Through my master's degree in Data Science at the University of Sydney, I expect to gain and improve competencies in data management, machine learning, AI-driven analytics, and automation, which are critical for executing my REAP. Specifically, I will develop technical proficiency in Python, SQL, and AI models, enabling me to design and implement PRIgpt: RAILytics, an AI-powered tool for competency gap analysis at DOTr-PRI. This master's program will enhance my data engineering and analytical skills, allowing me to build a structured competency database for RO personnel. Courses in big data analytics and predictive modeling will equip me with the ability to analyze workforce data, ensuring accurate training recommendations. Additionally, exposure to AI ethics and data privacy will strengthen my capability to manage sensitive personnel information securely and responsibly. By mastering these competencies, I will be able to modernize railway's workforce development system, transitioning from a manual training needs assessment to a data-driven, automated approach. Ultimately, my studies will provide the expertise required to develop scalable, AI-driven solutions that can support the future expansion of competency-based training which can be adopted to all railway operator personnel.

III. IMPLEMENTABILITY OF THE REAP. The REAP should be fully implemented within two years from when a scholar returns from Australia. The applicant should clearly and succinctly explain how he/she will implement the REAP within the required timeframe and budget. (TIP: The scale of your REAP should be within your circle of influence and control to ensure doability.)

Measurability

How will you track the progress of the implementation of your REAP? What are your milestones? Divide your REAP activities and timeline into four (4) milestones: 25%, 50%, 75% and 100%.

Action Steps (What activities will contribute to the milestones)	Expected Output	Timeline
1. A Comprehensive Competency Database of PRI Personnel		
Stakeholder Engagement & Planning	PRI leadership, HR, and training teams engaged; framework for data collection defined	Months 1-4 (25%)
Data Collection & Population	50% of PRI personnel data collected and verified	Months 5-6 (50%)
Database Completion & Initial Analysis	90% of personnel competency records finalized, preliminary	Months 7-8 (75%)

	competency analysis begins	
Database Validation & System Readiness	Data fully validated and verified, ready for integration into PRIgpt: RAILytics	Months 9-10 (100%)
2. The PRIgpt: Railytics Tool for Competency Data Analysis		
AI Model Design & System Architecture	PRIgpt: RAILytics framework and AI model requirements established	Months 11-14 (25%)
Prototype Development & Testing	Initial PRIgpt: RAILytics prototype developed and tested on collected data	Months 15-17 (50%)
Pilot Deployment & Feedback Collection	AI tool tested with select PRI trainers and HR personnel, user feedback gathered	Months 18-20 (75%)
Full System Integration & Refinement	AI-generated insights integrated into PRI training planning; tool refined	Months 21-24 (100%)
3. The PRIgpt: RAILytics Guideline		
Guideline Structure & Initial Documentation	PRIgpt: RAILytics Guideline framework outlined, documenting the automation process	Months 9-10 (33%)
Refinement & Lessons Learned	Incorporate pilot testing results and system improvements into the guideline	Months 11-20 (66%)
Finalization & Institutionalization	Guideline completed and integrated into PRI's workforce development policies	Months 21-24 (100%)

Resources	To minimize costs, several strategies can be adopted by leveraging existing resources, open-source tools, and strategic partnerships. 1. **Personnel (Data Collection & Processing):** Leverage existing PRI personnel (HR & Training teams) to gather and validate data as part of their roles. 2. **Database Development & Cloud Storage:** Use Microsoft OneDrive, or a PRI-hosted intranet for data storage instead of paid cloud services. For structured databases, use Google Sheets or Airtable (free version) instead of SQL-based paid solutions. 3. **AI System Development & Training:** Use open-source AI models (e.g., Ollama, LangChain, Hugging Face models) instead of custom-built AI from scratch. Collaborate with local universities with AI expertise to get free consulting or student projects. 4. **Software & Tools (LLM, AI, Data Analytics):** Use free and open-source data analytics tools like Python (Pandas, TensorFlow), Power BI (free version), or Google Colab (free cloud computing). Explore government AI initiatives for possible tech support. 5. **Training & Capacity Building:** Conduct in-house training using PRI's existing trainers and resources. Utilize free online AI/data science courses from Coursera (Google/Microsoft-sponsored), edX, and YouTube. 6. **Pilot Deployment & System Integration:** Do a small-scale internal PRI trial using select departments before full rollout. Use PRI's existing IT infrastructure to integrate PRIgpt: RAILytics instead of purchasing new systems. 7. **Documentation & Institutionalization:** Use digital documentation (Google Docs, Notion, or Confluence) instead of printing. Assign documentation to existing PRI personnel to avoid hiring consultants.

REAP Start Date and End Date	Start Date: July 2027 (tentative) – After completion of master's degree program End Date: July 2029

RISKS AND MITIGATING MEASURE

Describe the risks that might impact on the success of the Action Plan and how you plan to manage these to ensure success. (Identify factors that will block/limit/slowdown accomplishment of intended results or lessen quality of outputs.)

RISK	*MITIGATING MEASURE*
Limited access to comprehensive personnel data	Secure management approval and ensure compliance with data privacy regulations to facilitate safe and ethical data collection.
Resistance to adopting AI-driven analytics	Organize capacity-building workshops to educate stakeholders on the benefits of PRIgpt: RAILytics and provide hands-on training for its effective use.

Technical challenges in AI model development and integration	Collaborate with data science experts and academic institutions to ensure a robust AI model. Utilize open-source tools and cloud-based solutions to reduce development costs and improve scalability.
Limited funding for development and implementation	Seek institutional support or partnerships to supplement financial resources. Optimize the use of existing digital infrastructure to minimize additional costs.
Sustainability of PRIgpt: RAILytics beyond initial implementation	Embed the tool into PRI's long-term competency management strategy, provide training to internal staff, and integrate it into organizational policies to ensure continued use.
Potential cybersecurity and data privacy concerns	Implement strict data protection protocols, comply with national cybersecurity standards, and conduct regular security audits to ensure data integrity.

Signed by:

Engr. Rhodolph C. Nullar
Name of Proponent

Signed by:

ANNELI R. LONTOC, *CESO I*
Undersecretary and OIC-Executive Director